



Yug Sarda

Roll No.: 2023UCP1663

B.Tech in Computer Science and Engineering

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EDUCATION

•Malaviya National Institute of Technology, Jaipur

2023 – 2027

B.Tech in Computer Science and Engineering

CGPA: 8.10/10

•Narayana Junior College

2021 – 2023

State Board (Intermediate)

Percentage: 96.8%

PERSONAL PROJECTS

•Autonomous Multi-Agent AI Research System

FastAPI, LangGraph, LlamaIndex, ChromaDB

- Architected and developed an end-to-end autonomous multi-agent AI research platform using LangGraph with a microservices architecture comprising API Gateway, Ingestion, Memory, and Query services for scalable and modular AI workflows.
- Built a Hybrid Retrieval-Augmented Generation (RAG) pipeline using ChromaDB, Hybrid Search, Multi-Query Retrieval, RAG Fusion, Maximal Marginal Relevance (MMR), and Cross-Encoder reranking to improve document recall, retrieval precision, and context quality.
- Implemented Redis Cache for low-latency retrieval and Redis Streams with Consumer Groups to enable reliable asynchronous communication and distributed task processing across microservices.
- Designed persistent conversational memory using PostgreSQL and integrated LangSmith, Prometheus, and Grafana for end-to-end tracing, observability, latency monitoring, and debugging across distributed services.

•Customer Churn Prediction System

Python, Pandas, Matplotlib

- Built a machine learning system to predict customer churn based on behavioral, demographic, and service-usage patterns.
- Performed extensive data preprocessing including missing-value imputation, categorical encoding, feature scaling, and outlier handling.
- Implemented and compared models such as Logistic Regression, Random Forest, and XGBoost.
- Evaluated the system using cross-validation, ROC-AUC, precision, recall, and confusion matrix for robust performance.

•Healthcare Document Intelligence System

FastAPI, Neo4j, ChromaDB, Streamlit

- Developed a multimodal healthcare document intelligence platform supporting PDF and medical image understanding using Gemini Vision for OCR and visual reasoning.
- Architected a Hybrid Graph RAG pipeline combining Neo4j Knowledge Graphs, semantic retrieval, and patient memory to deliver personalized and context-aware clinical question answering.
- Implemented fault-tolerant multi-agent workflows for document ingestion, entity extraction, graph construction, validation, and clinical reasoning with automatic recovery from pipeline failures.
- Architected RESTful APIs using FastAPI with a layered backend architecture, separating routing, business logic and services for improved scalability and maintainability.

TECHNICAL SKILLS AND INTERESTS

Languages: C, C++, Python, SQL

Frameworks/Libraries: Numpy, Pandas, Matplotlib, LangChain, LangGraph, FastAPI

Developer Tools: Git, Postman, VS Code, Jupyter Notebook, LangSmith, Prometheus, Grafana

Related Coursework: Machine Learning, Deep Learning, Object-Oriented Programming, DBMS, Operating Systems, Data Structures and Algorithms

ACHIEVEMENTS

Secured 99.7 percentile and AIR rank 3402 in JEE Main and qualified for JEE Advanced

Selected in Amazon ML Summer School among 60000 applicants

Solved over 500+ DSA problems on LeetCode and GFG

LeetCode Knight (Maximum Rating: 1903)